

Md. Jubayer Hossain

Visiting Scholar, Department of Public Health, Daffodil International University

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Education and Training

Master of Science in Microbiology

Jagannath University

Dhaka, Bangladesh

2019–2020

- “Medium of Instruction: English”
- “Capstone Project: Genomic surveillance, clinical epidemiology, and outbreak response frameworks for Chikungunya in Bangladesh”

Bachelor of Science in Microbiology

Jagannath University

Dhaka, Bangladesh

2016–2019

- “Medium of Instruction: English”
- “Capstone Project: State the Art of Microbial Genomics”

Certificate Program in Interdisciplinary Computational Biology

BRAC University

Dhaka, Bangladesh

January 2026

- Protein structure prediction and molecular visualization using AlphaFold2/3 and ChimeraX.
- Applied AI/ML docking protocols to PDB and UniProt structural data for predictive molecular modeling.

Specialized Training in AI for Public Health

Child Health Research Foundation (CHRF)

Dhaka, Bangladesh

Sep 2023

- Built predictive models and data pipelines for disease surveillance and vaccine impact assessment in Bangladesh.
- “Skills: Epidemiological Modeling, AI-based Surveillance, Time Series Analysis (LSTM, ARIMA), Medical Image Analysis”

Technical Skills

Programming Languages: Python, R, Julia, SQL, Bash, JavaScript; **Bioinformatics:** Bulk RNA-Seq, Single-Cell RNA-Seq, Spatial Transcriptomics, Multi-omics; **Workflow Management & Reproducibility:** Nextflow (DSL2), nf-core, Docker, Singularity, Git, GitHub; **AI in Biology:** Scikit-learn, PyTorch, RDkit, DeepChem; **HPC & Cluster Computing:** SLURM (sbatch, sarray), Nextflow DSL2 with SLURM executor for scalable reusable pipelines; **Cloud:** AWS S3, Google Cloud Storage for large genomic datasets

Academic Appointments

Visiting Scholar

Department of Public Health, Daffodil International University

Dhaka, Bangladesh

Sep 2025 – Present

- Collaborated with faculty members to develop new research protocols for molecular and computational biology.
- Mentored junior researchers and students on data analysis techniques and the development of bioinformatics workflows.

Faculty (Bioinformatics)

University of Dhaka (cBLAST)

Dhaka, Bangladesh

Aug 2023 – Present

- Taught Biomedical Machine Learning and RNA-seq analysis to 100+ trainees across universities in Bangladesh.
- Directed student research projects applying deep learning to omics data.

Program Lead & Mentor

“Bioinformatics Mentorship Program (BMP) | mdjubayerhossain.com/bmp”

Dhaka, Bangladesh

2026 – Present

- Leads a 12-week intensive program training early-career researchers in RNA-seq analysis across cancer genomics and neurogenomics tracks.
- Mentees produce manuscript-ready, reproducible research outputs progressing to peer-reviewed publication.

Program Lead & Mentor

"GSA Bioinformatics Internship | gsabioinfinternship.owlstown.net"

Dhaka, Bangladesh

July 2025 – Present

- Organized in partnership with GNOBB, ASI School of Life, and SPSB bridging theoretical education and real-world bioinformatics research.
- Interns access the GNOBB-ASI Bioinformatics Hub to conduct RNA-Seq, multi-omics, and systems biology analyses.

Teaching Assistant

Department of Microbiology, Jagannath University

Dhaka, Bangladesh

Oct 2022 – Dec 2022

- Assisted in research methodology training covering experimental design and statistical analysis in public health microbiology.
- Supported departmental surveillance data processing and reviewed student research proposals.

Selected Publications

1. Ahmed, M. Z., Billah, M. M., Ferdous, J., & **Hossain, M. J.*** (2025). Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors. *Scientific Reports*, 15, 14405. <https://doi.org/10.1038/s41598-025-88339-z>
2. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & **Hossain, M. J.*** (2025). Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer. *Scientific Reports*, 15, 34490. <https://doi.org/10.1038/s41598-025-13315-6>
3. Bari, S. M., Fuad, M., **Hossain, M. J.** et al. (2025). A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout. *BMC Genomics*, 26, 999. <https://doi.org/10.1186/s12864-025-12127-2>

Papers Under Review

1. Islam, M. W., Fariha, F. T. J., Ahmed, M. Z., Ferdous, J., Lota, H. B., Hossain, M. S., Kundu, P., Shahjahan, M., & **Hossain, M. J.*** (2025). "Bioinformatics-driven multi-omics profiling of CDK1 and CDK6 identifies prognostic and therapeutic roles in breast cancer". *Under Review*.
2. Kundu, P., Shahariar, M., Munny, F. A., Riya, A. T., Lota, H. B., Ahmed, M. Z., Meme, M. A., Kona, U. M. P., Mahmuda, F., Chowdhury, S. S., Sraboni, S. A., & **Hossain, M. J.** (2026). Computational identification of *Taxus baccata*-derived phytochemicals targeting EZH2 to overcome therapeutic resistance in melanoma. *Research Square*. <https://doi.org/10.21203/rs.3.rs-8494315/v1>. *Under Review*.

Research in Progress

1. **Hossain, M. J.**, Shahjahan, M., & Fuad, M. (2025). Fezf2-mediated cortical development: Multi-omics analysis. *Working Paper/GitHub*. <https://github.com/hossainlab/fezf2-multiomics>

Analyzed ~125,000 cells across 23 scRNA-seq and 3 scATAC-seq samples spanning the full E10–P4 cortical developmental timeline under wild-type, heterozygous, and knockout conditions. Multi-omics integration with peak-to-gene linkage reveals Fezf2 as a master regulator of corticofugal neuron specification, with 7,248 genes showing a measurable linear dose-response to Fezf2 dosage and a critical intervention window at E13–E15.

2. **Hossain, M. J.**, Shahjahan, M., & Fuad, M. (2026). Pan-Cancer Tumor-Associated Macrophage Reprogramming Atlas and Immunotherapy Response Prediction. *Working Paper/GitHub*. <https://github.com/hossainlab/TAM-Atlas>

Integrated ~640,000 single cells from 8 published scRNA-seq datasets spanning lung, breast, liver, colorectal, melanoma, BCC, ovarian, and prostate cancers into a unified pan-cancer macrophage atlas. scVI (30 latent dimensions) was selected over conventional batch-correction methods to preserve the subtle transcriptional signals that define 6 biologically distinct TAM functional states; GRNBoost2 TF regulon reconstruction (MAFB, HIF1A) and pseudobulk differential expression link specific TAM programs to immune checkpoint blockade response.

Software and Pipelines

1. **metaXpress** (R/Bioconductor Package). End-to-end package for bulk RNA-seq meta-analysis integrating QC, differential expression (DESeq2, edgeR, limma-voom), and six meta-analysis approaches (Fisher's, Stouffer's, fixed/random effects). *GitHub*: <https://github.com/hossainlab/metaXpress>
2. **nf-core-rnaseqmeta** (Nextflow Pipeline). Reproducible pipeline for bulk RNA-seq meta-analysis across multiple studies with STAR/Salmon alignment, batch correction (ComBat-seq, SVA), and pathway enrichment. *GitHub*: <https://github.com/hossainlab/nf-core-rnaseqmeta>
3. **Salmon RNA-Seq Quantification Pipeline** (R/Shell Pipeline). End-to-end bulk RNA-seq quantification pipeline using Salmon and SRA-Tools for ultra-fast transcript-level abundance estimation. *GitHub*: <https://github.com/hossainlab/salmon-rnaseq>
4. **Single-Cell Analysis Workflow** (Bash, Python, and Workflow). End-to-end single-cell RNA-seq analysis workflow covering preprocessing, quality control, clustering, and cell type annotation. *GitHub*: <https://github.com/hossainlab/sc-workflow>

Leadership Experience

Founder, Bioinformatics Scientist & CEO

Dhaka, Bangladesh

“DeepBio Limited | deepbioltd.com”

Jan 2025 – Present

- Lead a team of 13 researchers executing scRNA-seq and bulk RNA-seq analysis pipelines for cancer genomics and transcriptomics projects.
- Develop and benchmark scalable bioinformatics workflows for biomarker discovery in cancer and neurodegenerative diseases.

Founder & Lead Instructor

Dhaka, Bangladesh

“DeepBio Academy | deepbioacademy.com”

June 2020 – Present

- Designed bioinformatics curriculum specializing in bulk and scRNA-seq workflows, multi-omics integration, and deep learning for biological applications.
- Mentored undergraduate researchers in building computational portfolios emphasizing open-science standards and version control.

Founder & Capacity Building Director

Dhaka, Bangladesh

“CHIRAL Bangladesh (Non-profit Research Institute) | deepbioltd.com/labs”

June 2020 – Present

- Trained 3,000+ early-career researchers in data science, research methodology, and computational biology workflows.
- Supervised 15+ manuscripts through analytical development and peer-review preparation in bioinformatics and global health.

Community Engagement

Nextflow Ambassador

Barcelona, Spain

Seqera

2025 – Present

- Advocated for the adoption of Nextflow-based DSL2 pipelines to ensure reproducibility and scalability in high-throughput single-cell and bulk RNA-seq workflows.
- Facilitated technical workshops on workflow orchestration, training researchers to manage complex multi-omic datasets with high analytical rigor.

GBD Senior Collaborator

Washington, USA

Institute for Health Metrics and Evaluation

2025 – Present

- Contributed advanced statistical modeling and analytical validation to Global Burden of Disease (GBD) studies published in high-impact journals like The Lancet.
- Collaborated with international research cohorts to refine statistical methodologies for assessing disease prevalence.

References

Dr. Syeda Tasneem Towhid

Associate Professor

Dr. Md. Salequl Islam

Professor

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